Salivary glands

Parotid, submandibular and sublingual glands: capsules, lobes, ducts, relations, nerves and the facial nerve landmarks.

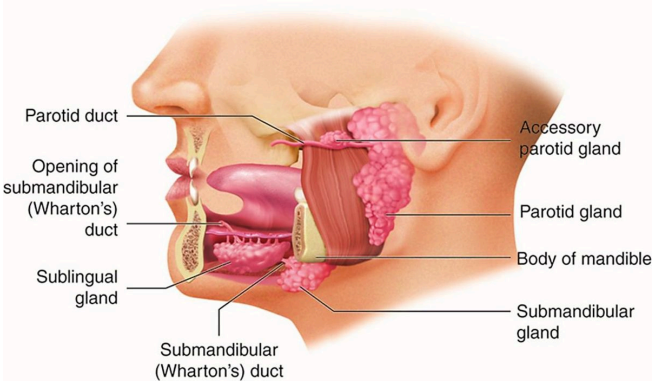
1. Overview and structure
2. Parotid: position, capsule and shape
3. Structures inside and around the parotid
4. Parotid duct and nerves
5. Facial nerve landmarks at parotidectomy
6. Submandibular gland
7. Sublingual and minor glands
8. On imaging
9. In theatre

01 Overview and structure

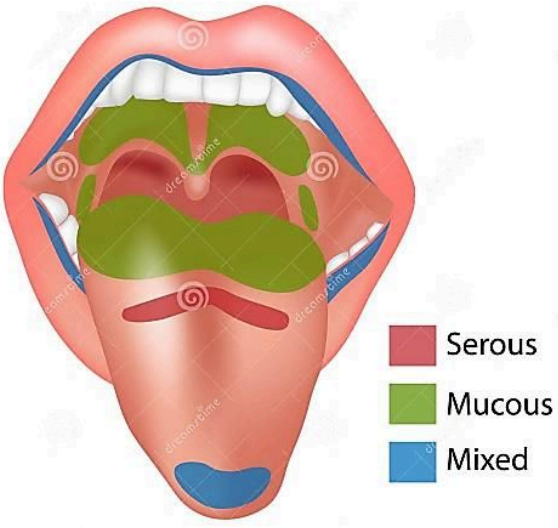
- **Major glands:** parotid, submandibular, sublingual. **Minor glands:** about 450 (ENT Steps) to 600–1000 (most textbooks) in the lips, cheeks, palate (most), floor of mouth, retromolar area, also in the sinuses, pharynx and larynx.
- Acini drain through **intercalated → striated → excretory ducts**; myoepithelial cells squeeze acini and intercalated ducts.
- **Parotid:** mainly serous. **Submandibular:** mixed. **Sublingual and minor glands:** mainly mucous.
- At rest the **submandibular gland gives most saliva** (about 70%); on stimulation the **parotid** takes over.

CLINICAL PEARL The smaller the gland, the more likely a tumour in it is malignant (parotid about 20%, submandibular about 50%, minor glands about 70–80%).

SOURCES DIFFER SUBLINGUAL GLAND: ENT Steps calls it the largest minor salivary gland; Doss and Gray's count it as a major gland. **NUMBER OF MINOR GLANDS:** ENT Steps 450; most texts 600–1000.



Major salivary glands and their ducts
ENT Steps Anatomy, p. 176



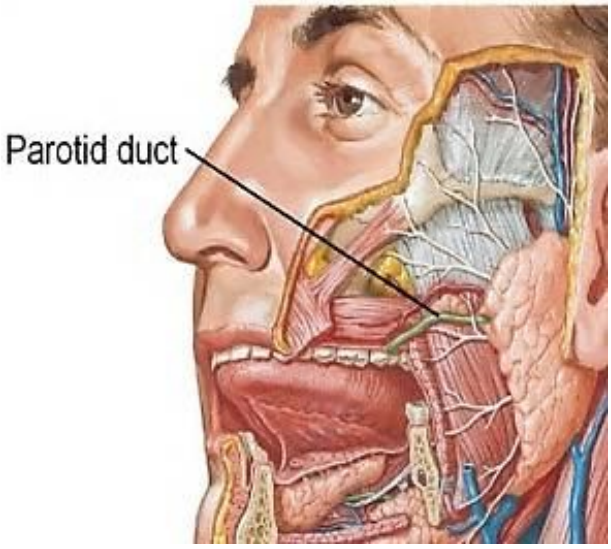
Distribution of serous and mucous minor salivary glands
ENT Steps Anatomy, p. 193

02 Parotid: position, capsule and shape

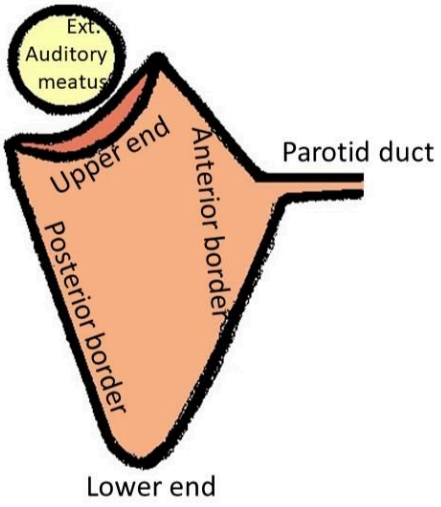
- Largest gland (about 25 g), in the space bounded by the **ramus, the EAC and the mastoid**; reaches the styloid process medially.

- **True capsule** from the gland's own stroma; **false capsule (parotid fascia)** from the investing layer of deep cervical fascia: thick superficial lamina to the zygomatic arch, thin deep lamina to the styloid and tympanic plate, forming the **stylomandibular ligament**.
- Shape: inverted three-sided pyramid. Apex below over the posterior belly of digastric; base above against the EAC and TMJ. Surfaces: superficial, anteromedial, posteromedial.
- The **facial nerve plane** divides it into a large **superficial lobe** and a **deep lobe**; an accessory lobe may lie on the masseter.

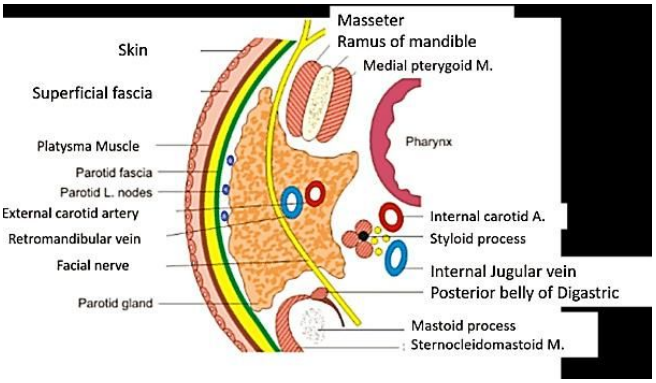
> **CLINICAL PEARL** Parotitis is very painful because the unyielding fascia is innervated by the great auricular nerve; pain worsens at meals.



Parotid gland and Stensen's duct
ENT Steps Anatomy, p. 181



Parotid gland: shape and surfaces
Dr. Sameh Doss handout, p. 72



Parotid in cross-section with its relations
ENT Steps Anatomy, p. 182

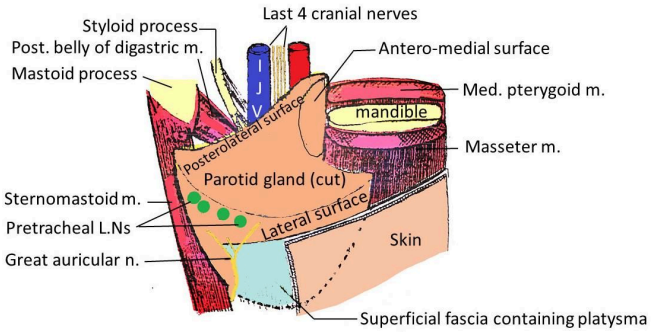
03 Structures inside and around the parotid

- **Within the gland, superficial to deep:** facial nerve and branches, **retromandibular vein**, **external carotid artery**, which gives the posterior auricular artery and divides into its two terminal branches, the superficial temporal and maxillary arteries. Also intraparotid lymph nodes.
- **Posteromedial surface:** mastoid, SCM, posterior belly of digastric, the styloid and its muscles; the styloid separates the gland from the ICA, IJV and IX–XII.
- **Anteromedial surface:** ramus, masseter, medial pterygoid, TMJ.

- **At the anterior border, top to bottom:** zygomatic branch, transverse facial artery, upper buccal branch, **parotid duct**, lower buccal branch, marginal mandibular branch.
- **At the apex:** cervical branch and the divisions of the retromandibular vein. **At the base:** temporal branch, superficial temporal vessels, auriculotemporal nerve.

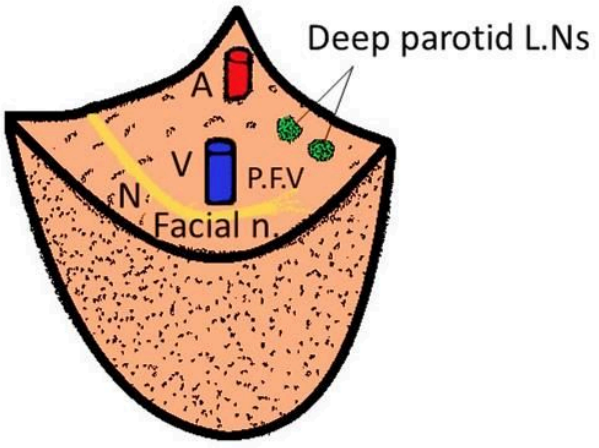
> **CLINICAL PEARL** The parotid is the only salivary gland with lymph nodes inside its capsule, because it encapsulates late in development.

> **SOURCES DIFFER ANTERIOR BORDER:** ENT Steps lists zygomatic, transverse facial artery, upper buccal, duct, lower buccal, marginal mandibular. Doss lists zygomatic branches, transverse facial artery, duct, buccal branch.



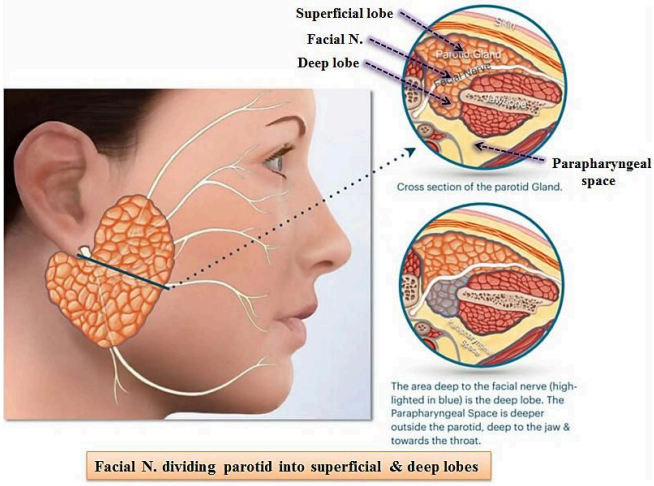
Relations of the parotid gland

Dr. Sameh Doss handout, p. 73



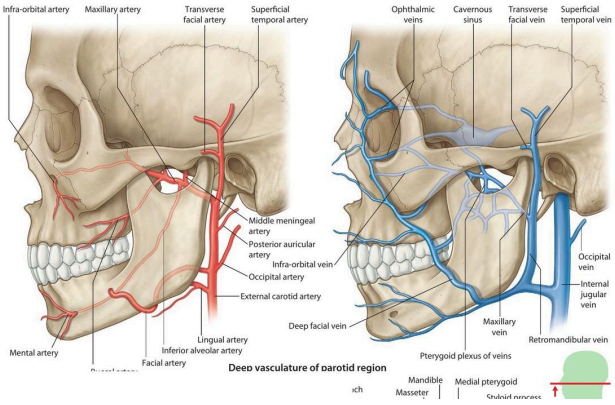
Structures within the parotid: facial nerve, vein, artery

Dr. Sameh Doss handout, p. 74



Facial nerve dividing the parotid into superficial and deep lobes

ENT Steps Anatomy, p. 181



Deep vasculature of the parotid region: arteries and veins

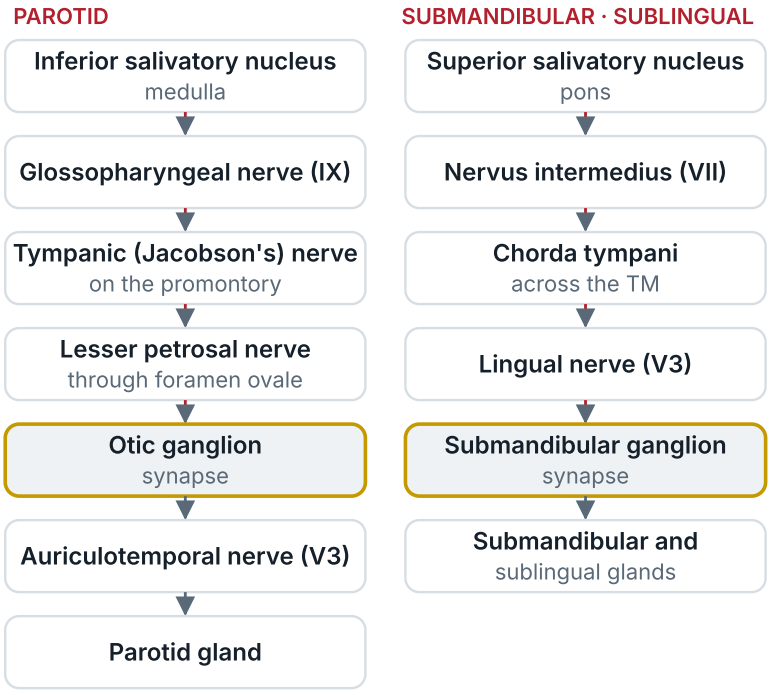
Gray's Atlas of Anatomy, 2nd ed., p. 507

04 Parotid duct and nerves

- **Stensen's duct:** about 5 cm long, leaves the anterior border a finger-breadth below the zygomatic arch, crosses the masseter, then pierces the **buccal fat, buccopharyngeal fascia and buccinator** to open opposite the **upper second molar**.
- Surface marking: middle third of a line from the tragus to the midpoint of the upper lip.

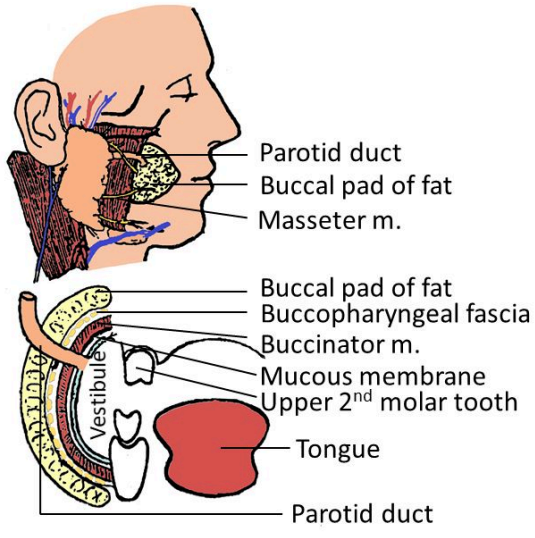
- **Secretomotor:** inferior salivatory nucleus → IX → tympanic (Jacobson's) nerve → lesser petrosal → otic ganglion → **auriculotemporal nerve**.
- **Sensory:** great auricular nerve (capsule), auriculotemporal nerve.

> **CLINICAL PEARL** Frey syndrome: after parotidectomy, cut secretomotor fibres regrow into the sweat glands of the skin, so the cheek sweats and flushes when eating.



Solid: preganglionic · dashed: postganglionic

Two parallel routes. Frey syndrome happens when the dashed parotid fibres in the auriculotemporal nerve regrow into skin sweat glands after parotidectomy.



Course of the parotid duct

Dr. Sameh Doss handout, p. 75

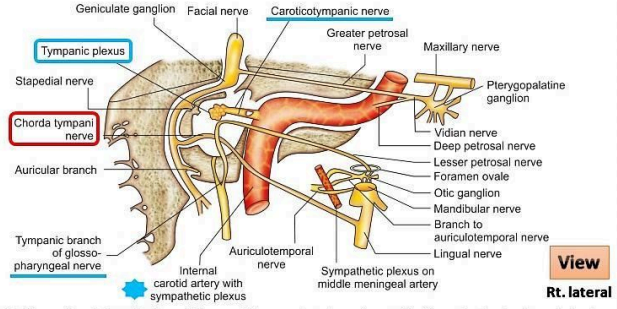
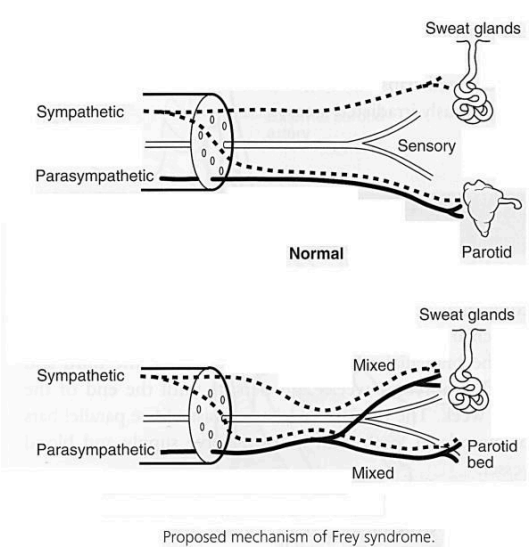


Fig. 25: Nerves in relation with the middle ear. Note secretomotor pathway of salivary, lacrimal and nasal glands

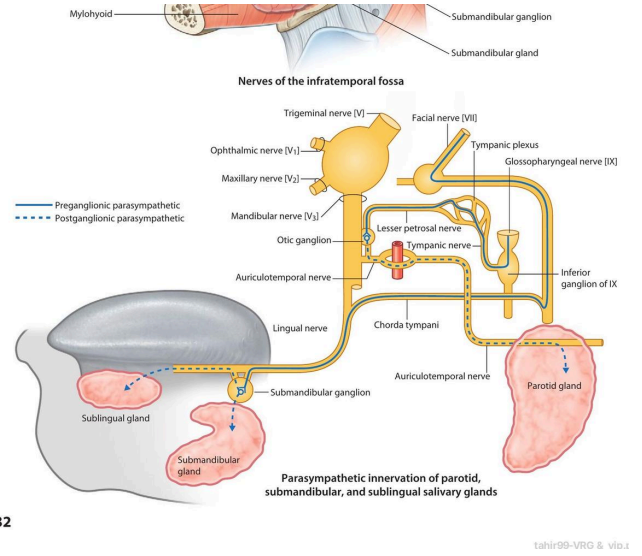
Nerves of the middle ear: tympanic plexus and petrosal nerves

ENT Steps Anatomy, p. 53



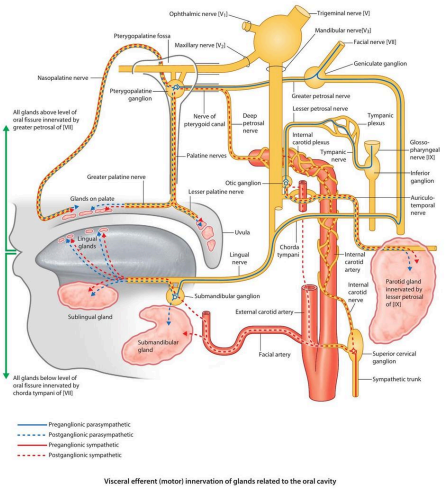
Mechanism of Frey syndrome

ENT Steps Anatomy, p. 188



Parasympathetic innervation of the parotid, submandibular and sublingual glands

Gray's Atlas of Anatomy, 2nd ed., p. 532

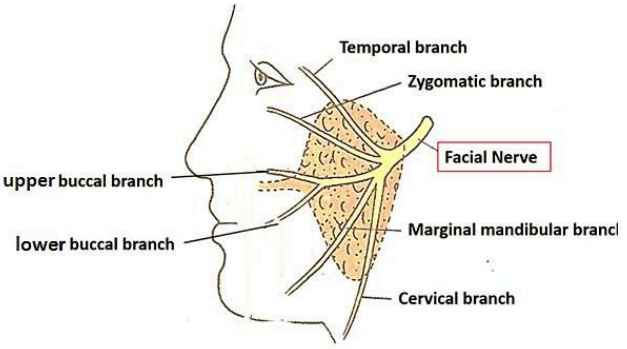


Autonomic innervation of the glands related to the oral cavity

Gray's Atlas of Anatomy, 2nd ed., p. 583

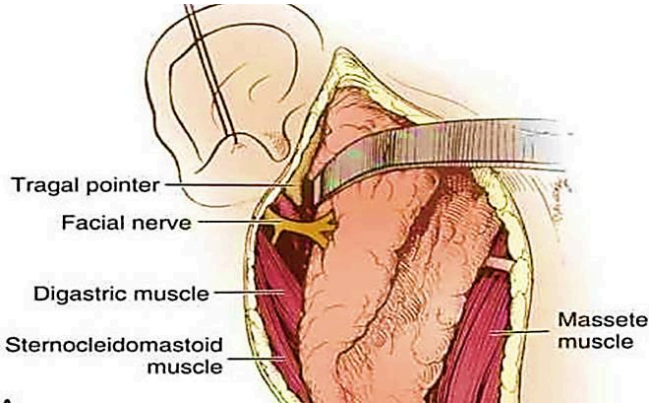
05 Facial nerve landmarks at parotidectomy

- Trunk leaves the stylomastoid foramen, enters the posteromedial surface and divides into **temporofacial** and **cervicofacial** divisions.
- **Tragal pointer:** nerve 1–1.5 cm deep and anteroinferior.
- **Tympanomastoid suture:** nerve 6–8 mm deep to it.
- **Posterior belly of digastric:** nerve just above its upper border at the mastoid.
- **Peripheral branches:** temporal branch crosses the zygomatic arch about halfway between the tragus and lateral canthus; buccal branch about 1 cm above the duct; marginal mandibular branch superficial to the facial vessels.



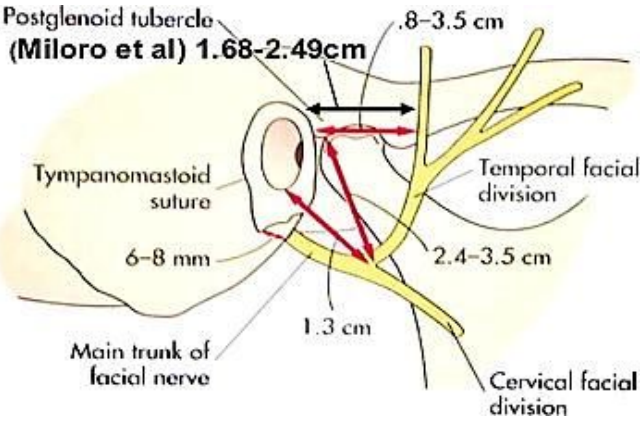
Branches of the facial nerve in the face

ENT Steps Anatomy, p. 183



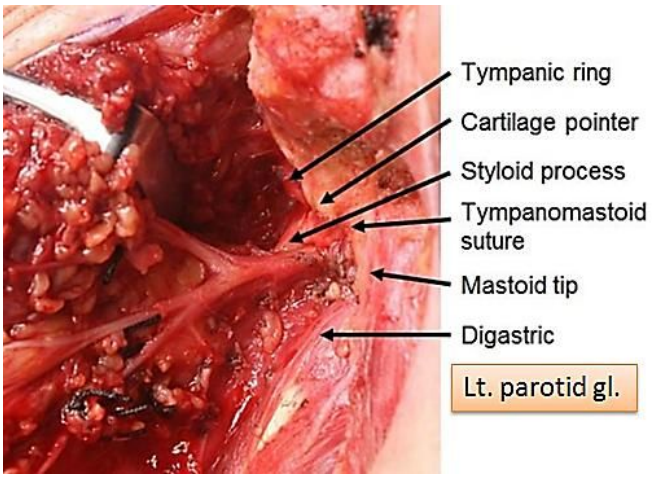
Landmarks for the facial nerve trunk: tragal pointer, digastric, mastoid

ENT Steps Anatomy, p. 185



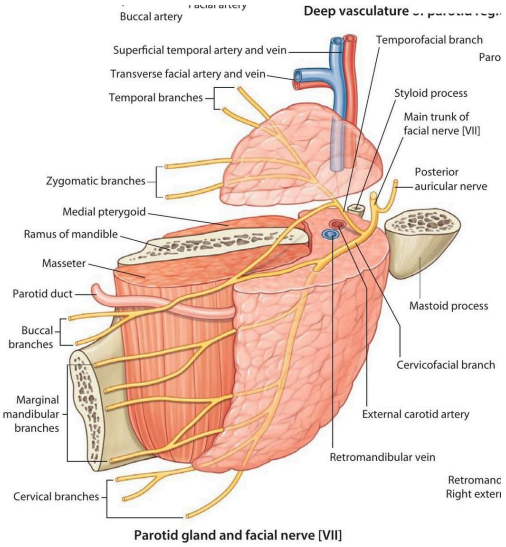
Distances from landmarks to the facial nerve trunk

ENT Steps Anatomy, p. 186



Surgical landmarks of the facial nerve at parotidectomy

ENT Steps Anatomy, p. 187



Parotid gland and facial nerve [VII]

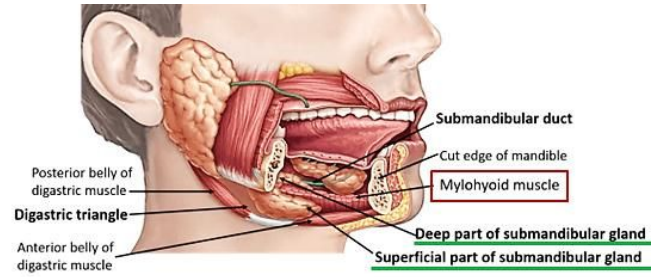
Gray's Atlas of Anatomy, 2nd ed., p. 507

06 Submandibular gland

- Fills the submandibular triangle. **Superficial part** (larger) on the mylohyoid; **deep part** on the hyoglossus; the two are continuous round the **posterior border of mylohyoid**.

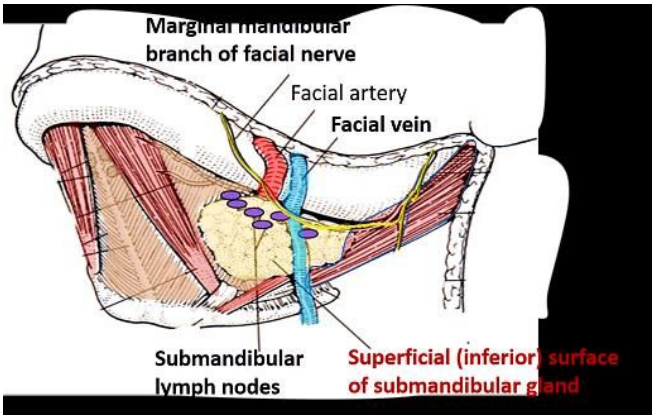
- Capsule from the investing layer of deep cervical fascia. **Facial vein** runs over the gland; the **facial artery** grooves its deep and upper surfaces.
- **Wharton's duct** (about 5 cm) leaves the deep part, runs forward on the hyoglossus between the **lingual nerve above** and **hypoglossal nerve below**, and opens at the **sublingual papilla** beside the frenulum.
- The **lingual nerve hooks under the duct** (lateral, then below, then medial).

> **CLINICAL PEARL** Excision: the **MARGINAL MANDIBULAR NERVE** lies superficial to the facial vessels just below the mandible. Hayes Martin manoeuvre: divide the facial vein low and lift it up to protect the nerve. Then look out for the lingual and hypoglossal nerves deep to the gland.



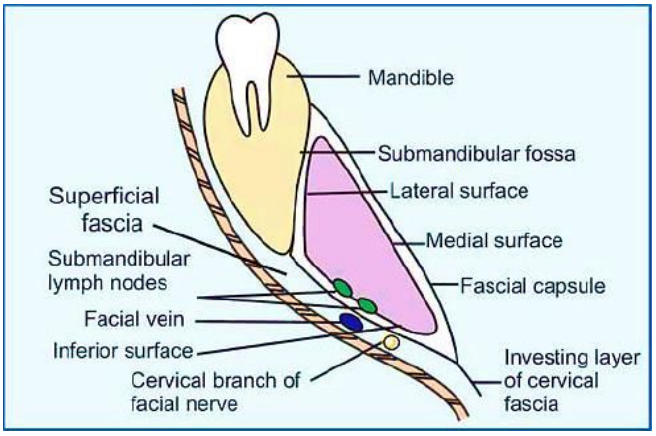
Submandibular gland around the mylohyoid

ENT Steps Anatomy, p. 189



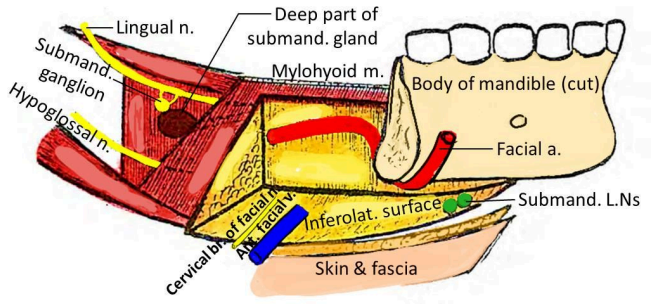
Marginal mandibular nerve and the submandibular gland

ENT Steps Anatomy, p. 190



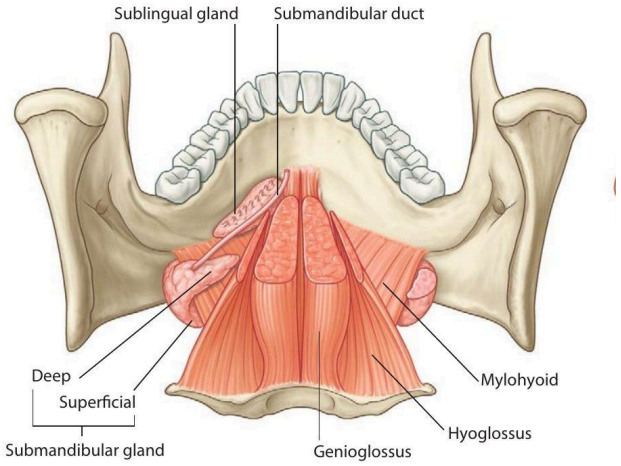
Fascial capsule of the submandibular gland

ENT Steps Anatomy, p. 190

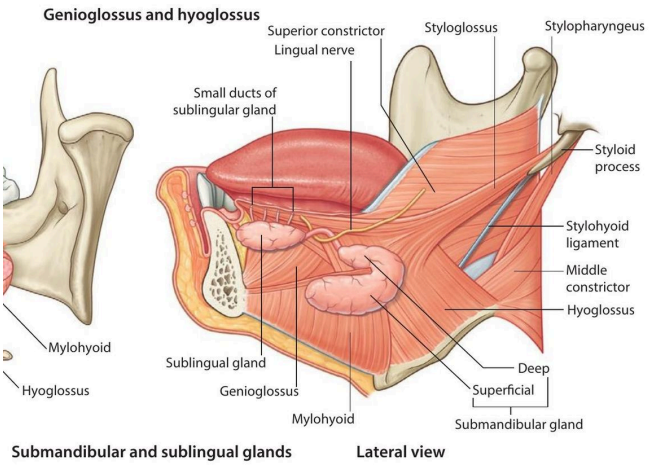


Submandibular gland: deep part and duct relations

Dr. Sameh Doss handout, p. 120



Submandibular gland around the mylohyoid (posterior view)
Gray's Atlas of Anatomy, 2nd ed., p. 578

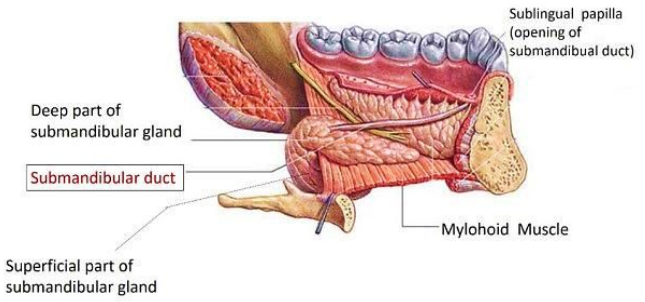


Submandibular and sublingual glands (lateral view)
Gray's Atlas of Anatomy, 2nd ed., p. 578

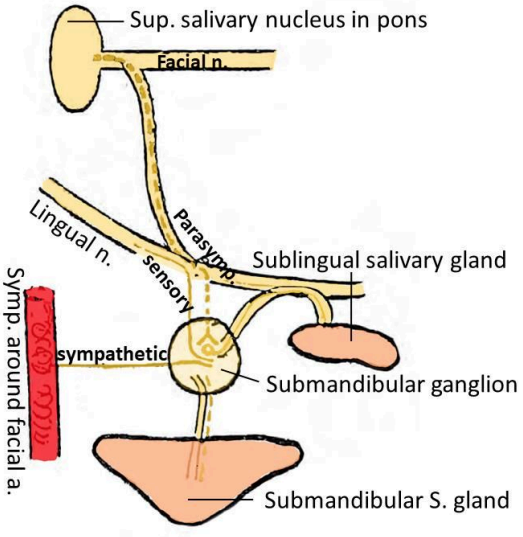
07 Sublingual and minor glands

- **Sublingual gland:** almond-shaped, in the front of the floor of the mouth between the mucosa, mylohyoid and mandible. Drains through about 10–20 **ducts of Rivinus** into the floor of the mouth, and some through **Bartholin's duct** into Wharton's duct.
- **Secretomotor to submandibular and sublingual:** superior salivatory nucleus → nervus intermedius → **chorda tympani** → lingual nerve → **submandibular ganglion**.
- Minor glands are mucous and contribute about 7–10% of saliva.

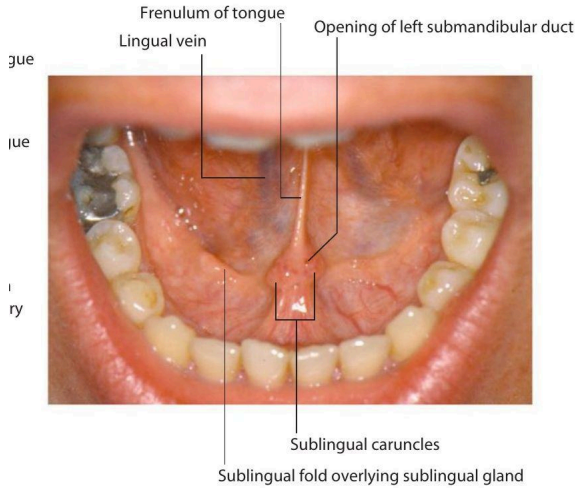
CLINICAL PEARL Ranula is a mucous extravasation cyst from the sublingual gland; a plunging ranula passes through or behind the mylohyoid into the neck.



Sublingual and submandibular glands with their ducts
ENT Steps Anatomy, p. 191



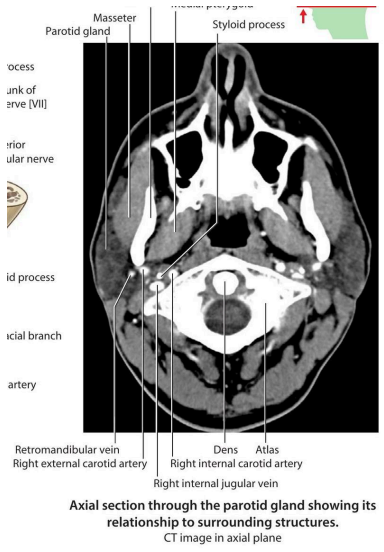
Secretomotor supply of the submandibular and sublingual glands
Dr. Sameh Doss handout, p. 121



Sublingual glands (anterosuperior view)
Gray's Atlas of Anatomy, 2nd ed., p. 582

08 Radiological anatomy: On imaging

- **Ultrasound** is first line: shows stones, cysts, solid lesions and nodes, and guides FNA.
- **MRI:** deep lobe extension, perineural spread along VII and V3, and malignancy features (low T2, ill-defined margins). Pleomorphic adenoma is very bright on T2.
- **Sialography and sialendoscopy** for ductal stones and strictures. Warthin's tumour is hot on technetium scanning.



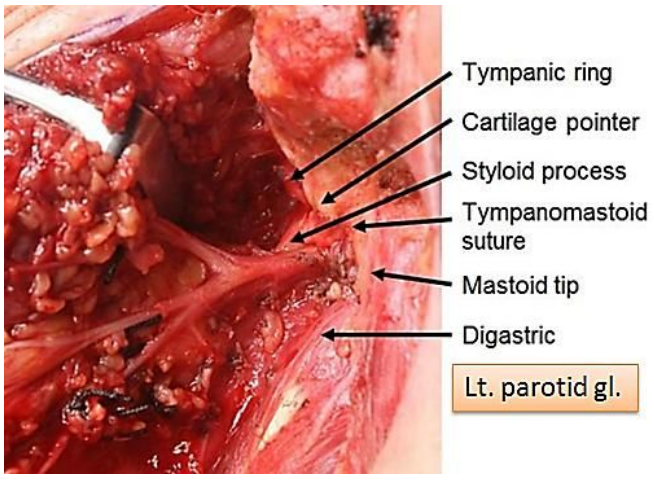
Axial CT through the parotid gland and its relations
Gray's Atlas of Anatomy, 2nd ed., p. 507

09 Surgical anatomy: In theatre

- **Superficial parotidectomy:** modified Blair (lazy-S) incision → skin flap above the parotid fascia → separate from the SCM and the cartilage of the EAC → identify the trunk with at least two landmarks → dissect along the nerve to remove the superficial lobe.
- **Extracapsular dissection** for small, mobile, superficial benign tumours.
- **Great auricular nerve:** keep its posterior branch if possible to reduce ear lobe numbness.

- **Submandibular gland excision:** incision two finger-breadths below the mandible, raise subplatysmal flap below the investing fascia over the gland (protecting the marginal mandibular nerve), ligate the facial artery and vein, identify the lingual nerve and divide the ganglion, then the duct; keep the hypoglossal nerve below.

> **CLINICAL PEARL** Enucleation of a pleomorphic adenoma leaves pseudopod extensions behind and causes multifocal recurrence.



Surgical landmarks of the facial nerve at parotidectomy
ENT Steps Anatomy, p. 187

Tables

Major salivary glands compared

	Parotid	Submandibular	Sublingual
Secretion	Serous	Mixed (mainly serous)	Mainly mucous
Share of resting saliva	~20–25%	~60–70%	~2–5% (minor glands ~7–10%)
Duct	Stensen's, opposite upper 2nd molar	Wharton's, at sublingual papilla	Ducts of Rivinus; Bartholin's duct
Secretomotor	IX → lesser petrosal → otic ganglion → auriculotemporal	VII → chorda tympani → lingual → submandibular ganglion	Same as submandibular
Nerves at risk in surgery	Facial nerve, great auricular	Marginal mandibular, lingual, hypoglossal	Lingual nerve, Wharton's duct
Malignancy rate of tumours	~20%	~50%	~70–80% (as minor glands)

Parasympathetic ganglia of the head

Ganglion	Preganglionic nerve	Hangs from	Supplies
Ciliary	Oculomotor (III)	Nasociliary (V1)	Sphincter pupillae, ciliary muscle
Pterygopalatine	Greater petrosal (VII) → vidian	Maxillary (V2)	Lacrimal, nasal, palatal and pharyngeal glands
Otic	Lesser petrosal (IX)	Mandibular (V3)	Parotid (via auriculotemporal)
Submandibular	Chorda tympani (VII) → lingual	Lingual (V3)	Submandibular and sublingual glands

Only the parasympathetic fibres synapse; sympathetic and sensory fibres pass through.

M Mnemonics

Parotid contents superficial to deep

NVA

Nerve (facial), Vein (retromandibular), Artery (external carotid)

Lingual nerve and Wharton's duct

Double cross

The lingual nerve crosses the duct twice: lateral, then below, then medial

Parotid anterior border, top to bottom

Zebras Trample Buttercups, Duck Beneath Mud

Zygomatic, Transverse facial artery, upper Buccal, Duct, lower Buccal, Marginal mandibular

Salivary malignancy

Smaller gland, bigger risk

Parotid ~20%, submandibular ~50%, minor glands ~70–80% malignant

Wharton's duct sandwich

Lingual above, hypoglossal below

On hyoglossus, the duct runs between the lingual nerve above and XII below

Q Rapid recall

1. **Where does the parotid duct open?** Opposite the upper second molar.
2. **Order of structures within the parotid, superficial to deep?** Facial nerve, retromandibular vein, external carotid artery.
3. **Which salivary gland produces most saliva at rest?** Submandibular (about 70%).
4. **What is Frey syndrome?** Gustatory sweating after parotidectomy from regrowth of parasympathetic fibres into sweat glands.
5. **Relation of the lingual nerve to Wharton's duct?** It hooks under the duct from lateral to medial.
6. **Nerve at risk with the incision for submandibular gland excision?** Marginal mandibular branch of the facial nerve.
7. **Which salivary gland contains lymph nodes?** Parotid (encapsulates late).
8. **Length of Stensen's duct?** About 5 cm.
9. **Surface marking of the parotid duct?** Middle third of a line from the tragus to the midpoint of the upper lip.
10. **Structures the parotid duct pierces?** Buccal fat, buccopharyngeal fascia, buccinator, buccal mucosa.
11. **What separates the parotid from the submandibular gland?** The stylomandibular ligament.
12. **Why is parotitis painful?** The unyielding parotid fascia stretches, stimulating the great auricular nerve.
13. **Structures emerging from the parotid apex?** Cervical branch of VII and the divisions of the retromandibular vein.
14. **Length of Wharton's duct?** About 5 cm.
15. **Where does the submandibular duct open?** Sublingual papilla beside the lingual frenulum.
16. **How does the sublingual gland drain?** About 10–20 ducts of Rivinus into the floor of mouth, some through Bartholin's duct into Wharton's duct.
17. **What is a plunging ranula?** A sublingual mucous extravasation cyst passing through or behind the mylohyoid into the neck.
18. **Hayes Martin manoeuvre?** Ligate the facial vein low and lift it up to protect the marginal mandibular nerve.
19. **Which salivary tumour is hot on technetium scan?** Warthin's tumour.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.